

Testing Brain-Inspired Mechanisms for Task-Incremental Continual Reinforcement Learning

A Controlled Ablation Study

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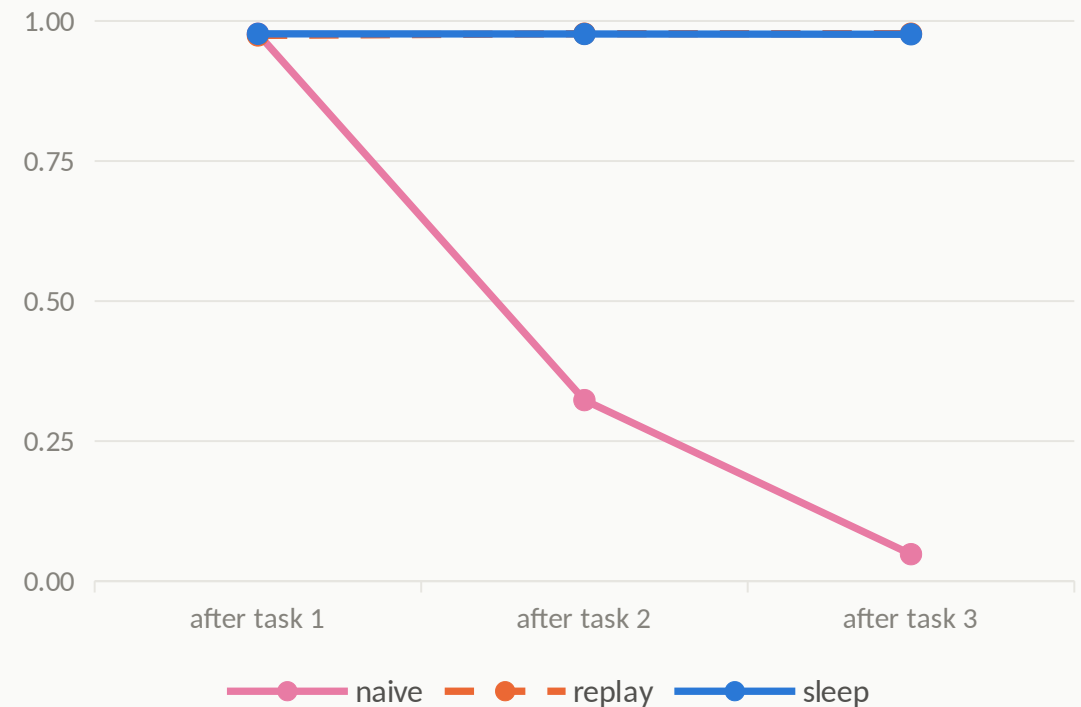
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Learning a new task erases the old one

- An agent learns tasks one after another: fetch the red ball, then the green key, then the blue box
- Plain fine-tuning overwrites what earlier tasks needed: catastrophic forgetting
- Three standard strategies: regularisation (EWC), replay of stored experience, capacity isolation (one network per task)
- Brain-inspired fixes (spikes, local plasticity, sleep) are popular, but are usually added all at once

Return on task 1 as later tasks are learned

CNN agent, 3 tasks, mean of 3 seeds; replay (dashed) and sleep overlap near 1.0



Which brain-inspired mechanisms actually help?

Switch each mechanism on alone and compare it with a control that is not brain-inspired but matches its resources. Six were tested; the seventh was inconclusive.

Spiking encoder

vs same-shape CNN, matched operations

STDP (local plasticity)

vs a random update of the same size

Homeostasis

vs the same agent without it

DFA: backprop-free encoder credit

vs ordinary backpropagation (heads exact in both)

Sleep-like consolidation

vs replay with an exactly matched replay budget

Shared trunk, per-task heads

vs one network per task, at equal total memory

Transformer memory (inconclusive)

vs no memory, and a stack of recent frames

Negative results count: the aim is to find out which mechanisms contribute, not to make all of them win.

Related work

● Continual RL: three strategies

Regularisation: EWC (Kirkpatrick et al., 2017); LwF distils old outputs, no stored data (Li & Hoiem, 2018)

Replay: CLEAR, a very strong baseline (Rolnick et al., 2019)

Isolation: PackNet, parameters per task (Mallya & Lazebnik, 2018)

● Sleep and consolidation

Sleep-like unsupervised replay recovers forgotten tasks in classifiers (Tadros et al., 2022)

Ours: offline replay + distillation in RL, compared with CLEAR-style replay

● Spiking networks for RL

Surrogate gradients make SNNs trainable (Neftci et al., 2019)

The surrogate slope matters more in RL (Van den Berghe et al., 2025)

Spiking decision transformers (Pandey & Biswas, 2025)

● Local learning rules

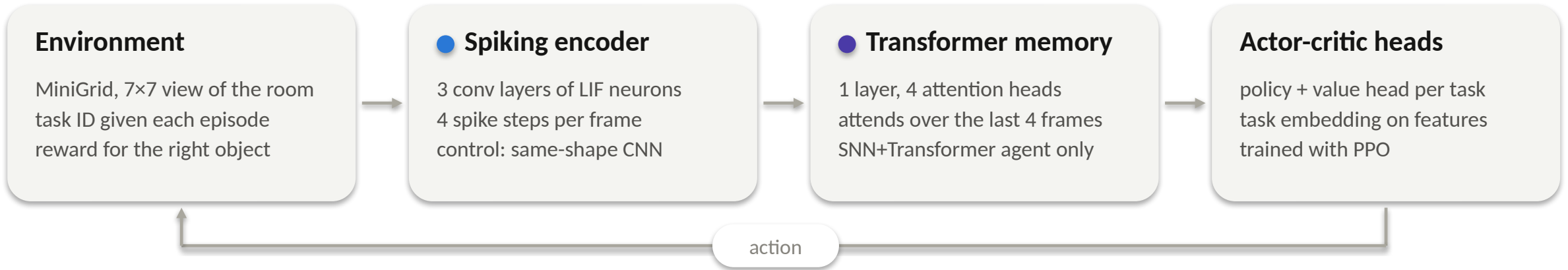
Three-factor, reward-modulated STDP (Frémaux & Gerstner, 2016)

Direct feedback alignment: no weight transport (Nøkland, 2016)

Homeostatic plasticity (Turrigiano & Nelson, 2004)

Gap: these mechanisms are usually tested together, against an unmodified baseline, so it is unclear which one did the work.

System architecture



Continual training of the sleep variant: wake and sleep phases alternate on the same shared trunk



Experimental setup and fairness controls

● Setup

- MiniGrid: fetch3 / fetch5 sequences (task ID given), DoorKey-6x6, Memory S7/S11/S13
- PPO, small agents (159,864–327,800 parameters)
- Metrics: final accuracy (ACC), forgetting, learning speed (AUC), solve rate

3–10

seeds per arm

289

PPO runs

110

single-thread CPU hours,
no GPU

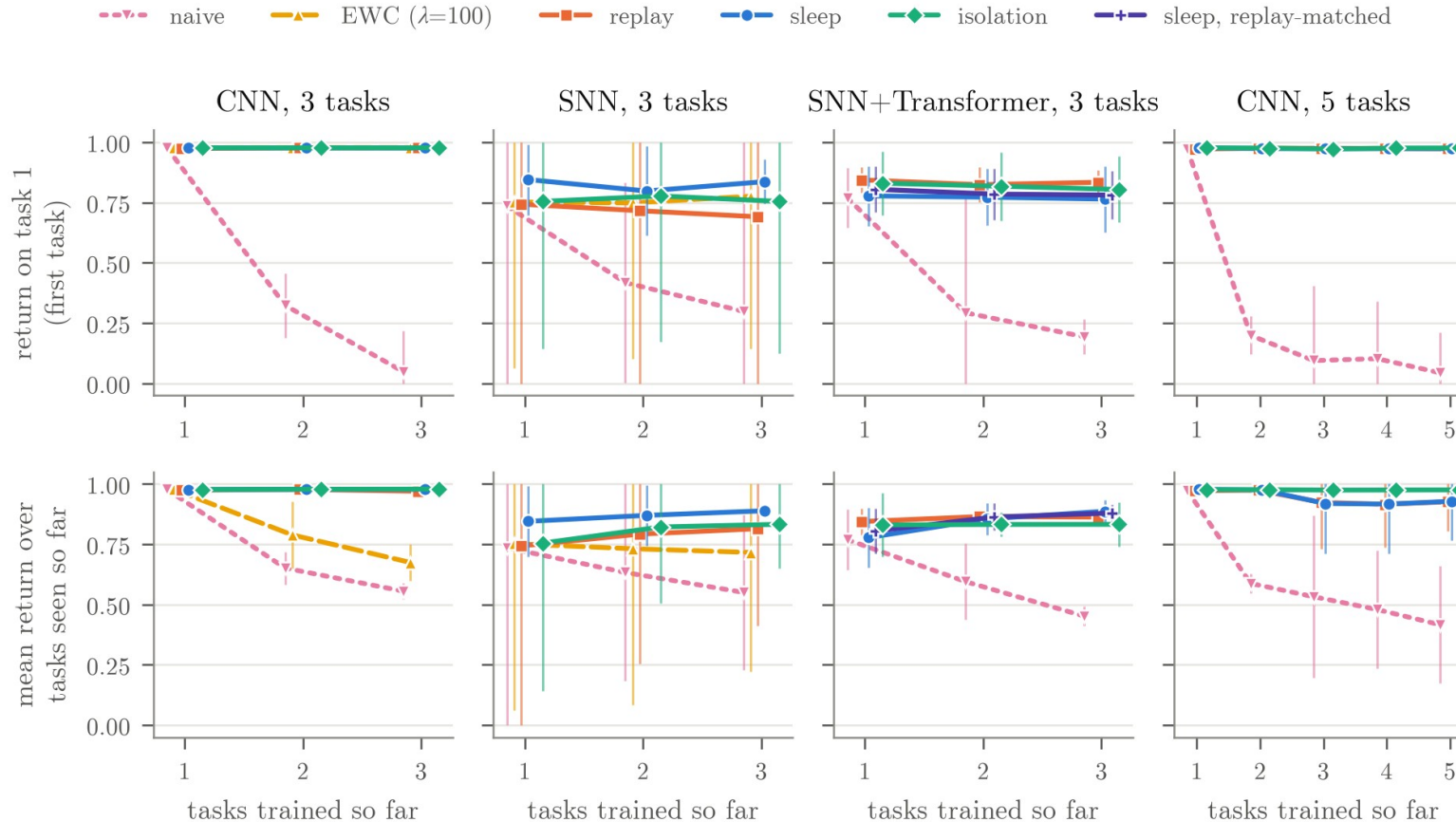
● Fairness controls

- Same frames, seeds and learning rate in every arm of a comparison
- STDP vs a random update of exactly the same size
- Sleep vs replay with the number of replayed samples matched exactly
- Memory counted as parameters plus replay buffer, not parameters alone
- Tests prospectively specified in a version-controlled experiment log; same seeds in every arm; Welch / Fisher; Holm; 95% CIs

No reliable improvement beyond the matched controls

MECHANISM	VERDICT	KEY NUMBER (95% CI)
Spiking encoder	ⓘ Fewer operations in an estimate; higher measured CPU cost	works at 16.6k ops/frame where the CNN needs 33k; measured CPU cost 5.2×
Local learning rule (STDP)	○ Not distinguishable from a same-size random update	tested rule vs random update: $\Delta\text{AUC} +0.033 [-0.260, +0.326]$, $p = 0.818$
Homeostasis, backprop-free encoder	✓ Faster learning, not more reliable	$\Delta\text{AUC} +0.326 [+0.101, +0.551]$, Holm $p = 0.017$; solved 7/10 vs 2/10 (n.s.)
Homeostasis, with backprop	○ Not significant after correction	$\Delta\text{AUC} +0.200 [+0.024, +0.377]$; Holm $p = 0.117$
Sleep-like consolidation	○ No detectable difference from replay at a matched budget	sleep – replay $\Delta\text{ACC} +0.017 [-0.026, +0.060]$, $p = 0.415$
Shared trunk with per-task heads	ⓘ Only with small buffers and full-width isolation	200 states/task: similar mean ACC in 0.89 vs 1.92 MB; post-freeze, quarter-width isolation beat it (0.972 vs 0.944)
Transformer working memory	? Inconclusive: benchmark validity not established	benchmark validity not established (no map both needed memory and was learnable)

Sleep and replay stop forgetting; no detectable sleep advantage at a matched budget



0.555

naive fine-tuning, final ACC (CNN, 3 tasks)

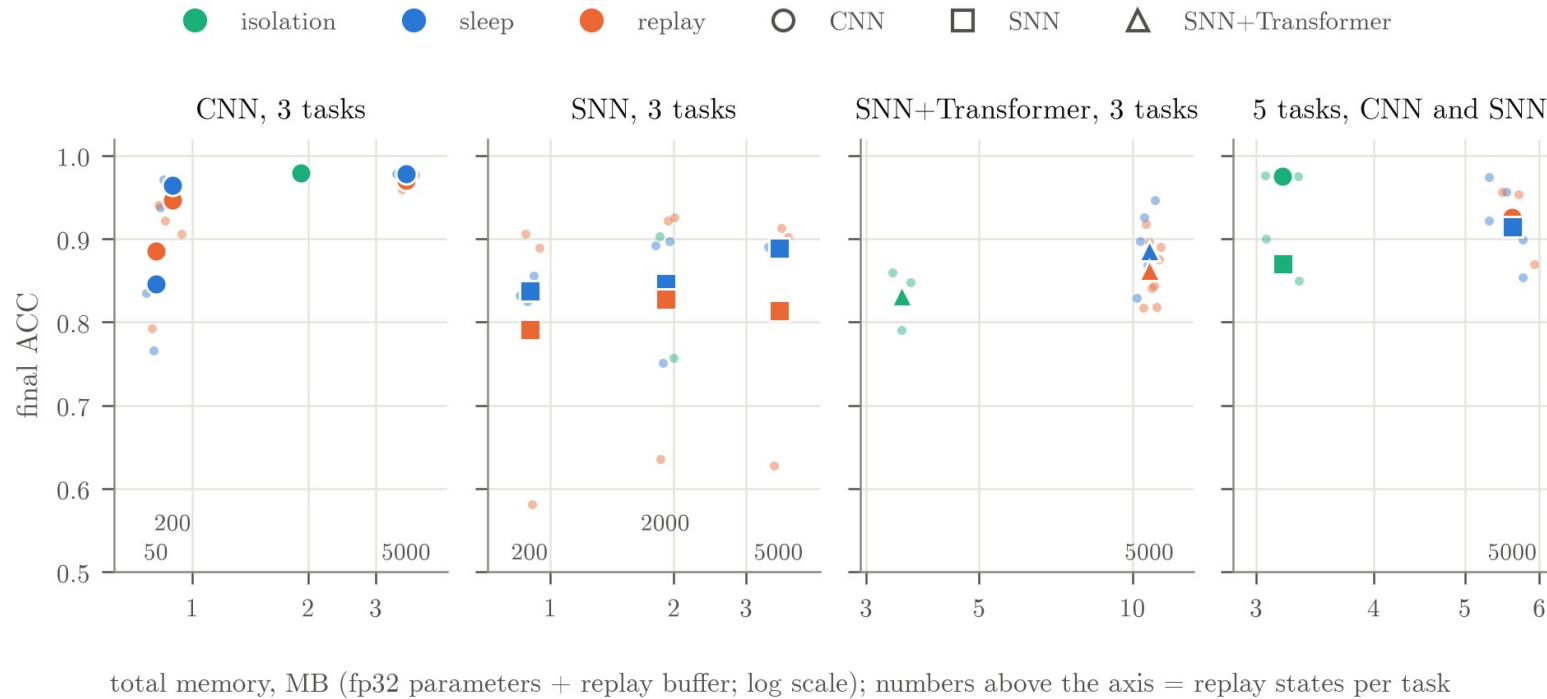
0.970 · 0.978

replay · sleep, final ACC

+0.017

sleep – replay at an exactly matched replay budget,
95% CI [-0.026, +0.060], $p = 0.415$

A shared trunk is competitive only with a small buffer and full-width isolation



Total memory = fp32 parameters + replay buffer. Numbers above the axis: replay states stored per task.

46%

of the memory: SNN sleep, 200 states/task, similar mean ACC to isolation (0.838 vs 0.833; 3 seeds, wide CI)

2.5–3.3×

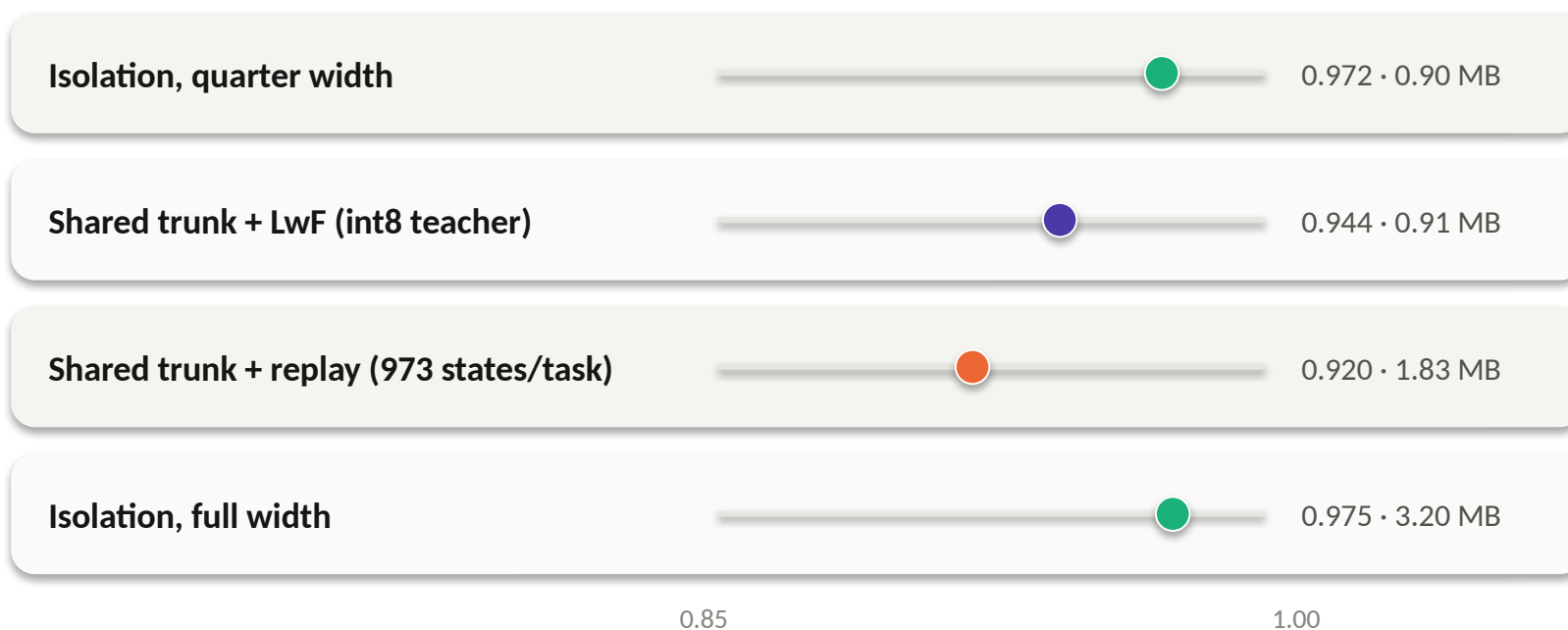
more accuracy per parameter for the shared trunk

1.8–2.9×

more accuracy per MB for isolation at the default 5,000 states/task

Post-freeze follow-up: narrower isolation networks win at equal memory

CNN, 5 tasks · 10 seeds per arm (seeds 201–210, the same in every arm) · confirmatory, post-freeze



Dots: mean final accuracy on a 0.85–1.00 scale. Memory: fp32 parameters plus stored states. The shared trunk was not shrunk the same way; the SNN case is untested.

+0.028

quarter-width isolation – shared trunk + LwF, at equal memory; 95% CI [+0.015, +0.041], Holm $p = 0.002$

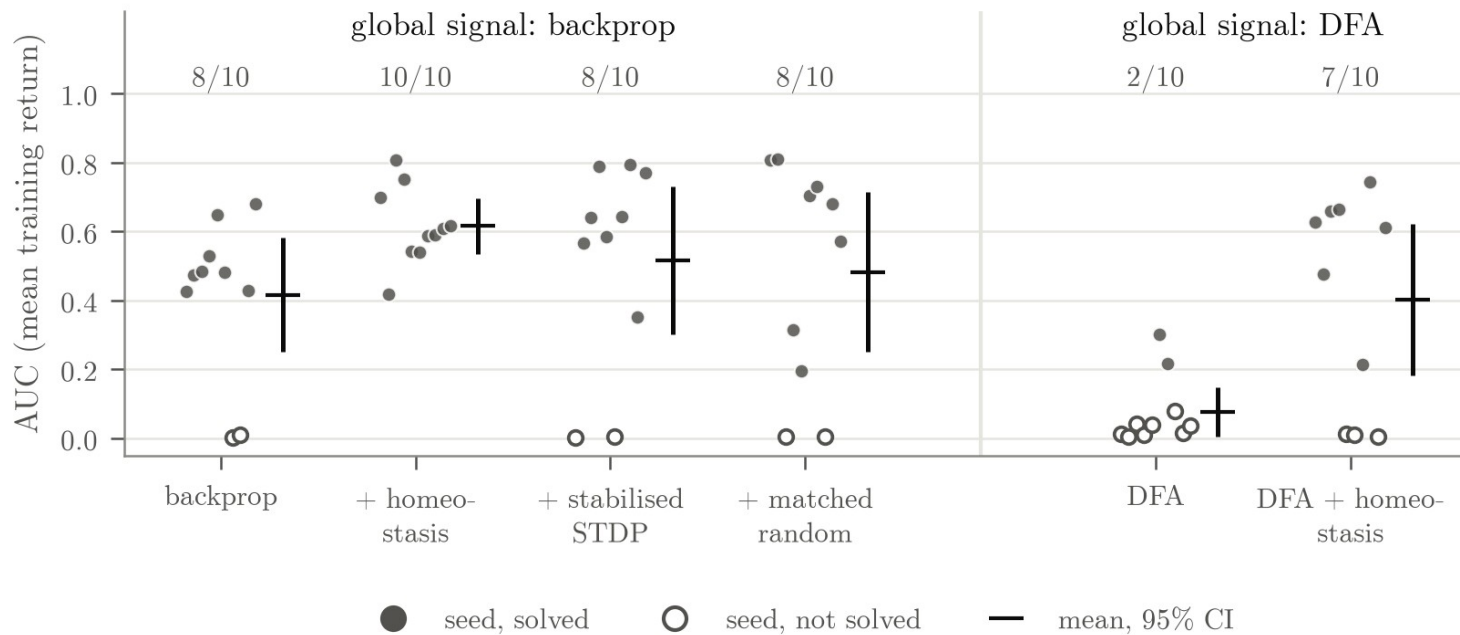
+0.052

vs shared trunk + replay with twice the memory; 95% CI [+0.015, +0.089], Holm $p = 0.011$

–0.004

vs full-width isolation at 3.6× the memory: no large difference

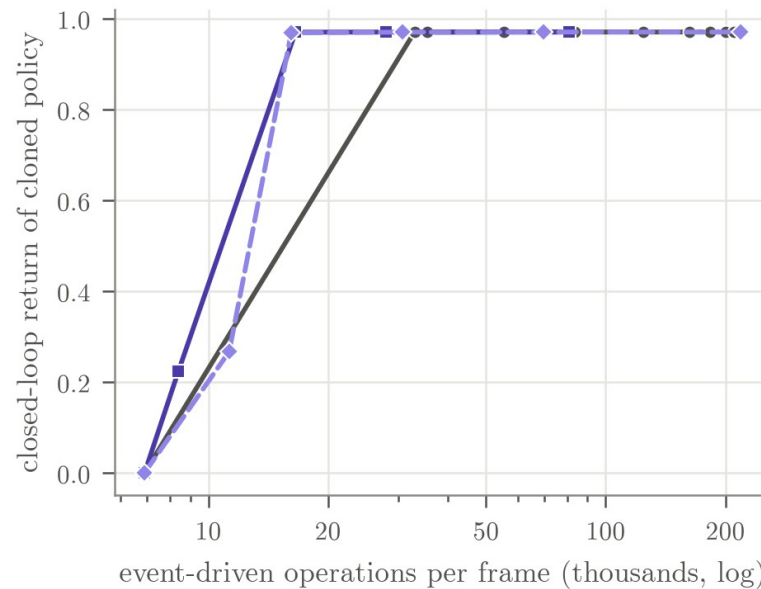
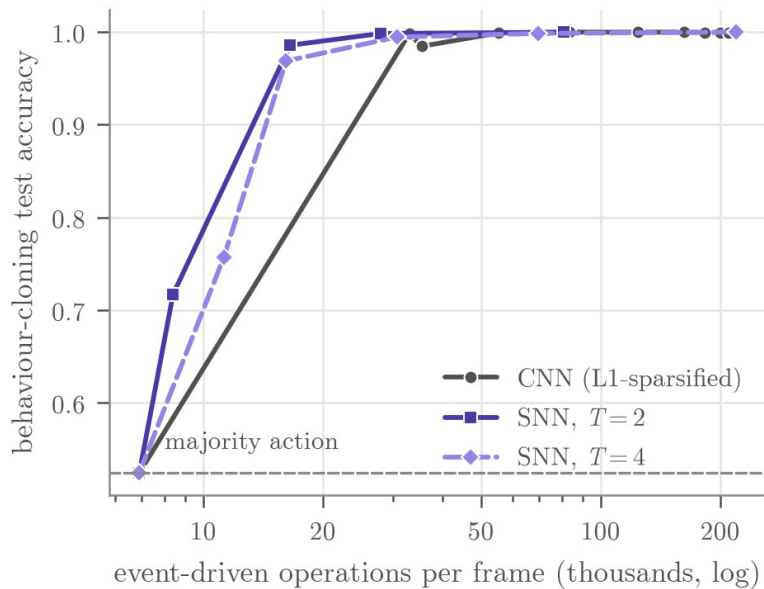
Tested STDP rule $\approx$ a same-size random update; homeostasis speeds up DFA



- **Tested STDP rule $\approx$ random update:** $\Delta\text{AUC} +0.033 [-0.260, +0.326]$, $p = 0.818$ (wide interval)
- **Unstabilised STDP hurts:** solved 0/3 vs 3/3 with backprop (runaway firing)
- **Homeostasis + backprop:** raw $p = 0.029$, Holm $p = 0.117$: not significant
- **Homeostasis + DFA (backprop-free encoder):** $\Delta\text{AUC} +0.326 [+0.101, +0.551]$, Holm $p = 0.017$; solved 7/10 vs 2/10 (Holm $p = 0.070$)

DoorKey-6x6, 10 seeds per arm (same seeds), confirmatory. Dots: seeds (filled = solved); bars: mean and 95% CI; numbers: seeds solved.

Spiking encoder: fewer operations, not more accuracy



16.6k

ops/frame: the SNN still works (accuracy 0.986)

33k

ops/frame: below this the sparsified CNN stops working

5.2×

measured cost in our implementation: CPU ms per training step on 256 observations, 1 thread (80 vs 16); energy is an estimated arithmetic proxy

Behaviour cloning of a DoorKey-6x6 teacher, activity penalty swept; 2 seeds per point. Operations = synaptic events (SNN) or non-zero multiply-accumulates (sparsified CNN).

What didn't survive controls, and why

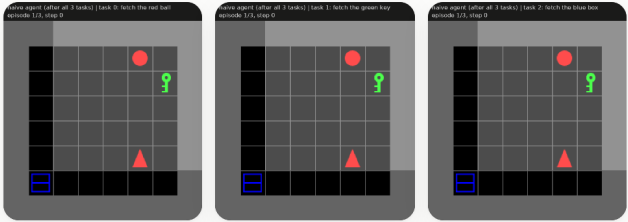
EARLIER FINDING	EVIDENCE THEN	AFTER THE CONTROL	CAUSE
Sleep beats replay (SNN+Transformer)	p = 0.072 with 3 seeds	tie at a matched replay budget: p = 0.415, 8 seeds	seeds; replay budget
Shared trunk far more efficient	2.5–3.3× per parameter	isolation 1.8–2.9× better per MB	memory accounting
Sleep far beats isolation, 5 tasks (SNN)	gap +0.303 [+0.148, +0.457]	gap +0.044 [−0.009, +0.098] at a fair budget	training budget
Homeostasis helps backprop	p = 0.088 (3 vs 5 seeds)	Holm p = 0.117 with 10 seeds	seeds; multiple tests
Homeostasis makes DFA work	solved 3/3 vs 1/5	7/10 vs 2/10, p = 0.070 (AUC effect survives)	seeds
Transformer helps on a memory task	S7 pilot 0.970 vs 0.465	no memory needed on S7 (memoryless CNN 0.915)	benchmark validity
Small buffer: shared trunk beats isolation	vs full-width isolation: 1.08 vs 0.51 ACC/MB (CNN)	quarter-width isolation 0.972 vs LwF 0.944 at 0.90 vs 0.91 MB	network width (post-freeze)

Effects shrank or disappeared once seeds, training budgets, replay budgets, memory accounting and network width were made fair; the homeostasis-DFA speed-up survived.

Live demo

Naive fine-tuning

seed 1 · final accuracy 0.541 · mean of 3 seeds 0.555



Task 1: red ball
✓ X X (1 of 3)

Task 2: green key
✓ ✓ X (2 of 3)

Task 3: blue box
✓ ✓ ✓ (3 of 3)

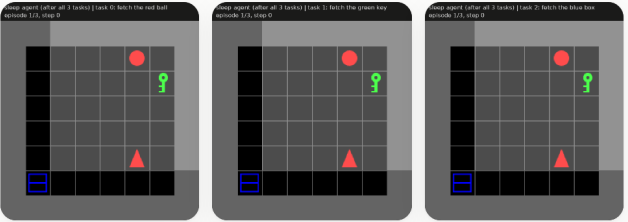
Task 1: the first clip happens to succeed (1 of 3 episodes here).
Over 100 evaluation episodes this agent's task-1 return is 0.126;
the matrix gives the real average.

	task 1	task 2	task 3
after task 1	0.977	0.275	0.119
after task 2	0.265	0.979	0.141
after task 3	0.126	0.542	0.956

Rows: after training task i. Columns: mean return on task j. The
bottom row is final performance; its mean is the final accuracy.

Sleep-like consolidation

seed 1 · final accuracy 0.978 · mean of 3 seeds 0.978



Task 1: red ball
✓ ✓ ✓ (3 of 3)

Task 2: green key
✓ ✓ ✓ (3 of 3)

Task 3: blue box
✓ ✓ ✓ (3 of 3)

9 of 9 episodes shown succeed, in line with the matrix: over 100
evaluation episodes per task, final returns are 0.978, 0.979,
0.979.

	task 1	task 2	task 3
after task 1	0.978	0.227	0.165
after task 2	0.979	0.980	0.080
after task 3	0.978	0.979	0.979

Rows: after training task i. Columns: mean return on task j. The
bottom row is final performance; its mean is the final accuracy.

1. Results dashboard

open [demo/index.html](#) (one file, works offline)
scorecard, agents, forgetting and memory charts

2. Run an agent on CPU

`python demo/live_episode.py`
 `--agent naive --task 0 --window`
then the same with `--agent sleep`
prints each step and the real accuracy matrix

Limitations

Task identity is given	task-incremental only; says nothing about task-agnostic learning
Small scale	MiniGrid, $\leq 327,800$ parameters, 3–5 tasks
Few seeds	many continual arms have 3; a tie rules out large effects only where the CI is narrow
Energy	an estimated arithmetic energy proxy from operation counts; no hardware power was measured
Isolation width	full width in the frozen study; the shared trunk was never shrunk the same way
Transformer memory	inconclusive: no benchmark both required memory and was learnable in budget
Informal prospective specification	a version-controlled experiment log, not a public registry

Next phase: scaling up

● Why

Every experiment ran on a 4-core CPU.

The bottleneck was CPU-bound environment stepping, not model size: it is what limited seeds, sequence length and training budgets.

● Plan

Port agent, mechanisms and baselines to JAX; run on cloud GPUs.

XLand-MiniGrid (Nikulin et al., 2024): GPU-native MiniGrid, reported to run millions of steps/s on one GPU, with a rule/goal system for many related tasks.

● Validation first

Before trusting anything new, replicate on XLand-MiniGrid's MiniGrid ports:

- sleep vs replay at a matched replay budget
- STDP vs a same-size random update
- the post-freeze isolation-width result (C4)

● Rough timeline

Port and replication: about 1–2 weeks

Scaled study: about 2–4 weeks

Then: paper revision and journal submission (TMLR first choice)

Next phase: open questions

● Questions the scaled study will address

- 10–50 related tasks; accuracy-vs-memory curves with every method shrunk along its own size setting
- 10–20 seeds per arm, power analysis and equivalence tests, so nulls become informative
- A learnable memory benchmark; then Transformer vs GRU vs frame stack, and a stateful SNN
- The SNN version of C4 at 450k frames per task
- Do the conclusions change with network size? A task-agnostic setting with observable goals

● Smaller follow-ups

- Replay-budget sweep: sleep vs interleaved replay (motivated by an exploratory result only)
- Sleep split: offline replay alone vs replay + self-distillation
- Plots of the STDP runaway mechanism
- Stronger CNN sparsifier (k-winners-take-all)
- Compressed or generative replay; a PackNet that can regrow capacity
- One standard benchmark, such as Continual World

Same methodology throughout: prospectively specified tests, fresh seeds for confirmation, matched resources, every failure logged.

CONCLUSION

No statistically reliable improvement beyond the matched controls

Given the seed counts, this rules out large effects rather than showing equivalence, and only where the intervals are narrow.

-  **Survived** homeostasis speeds up learning with backprop-free encoder credit assignment (DFA; heads exact), without making it reliable
-  **Conditional** shared trunk + small buffer $\approx$ full-width isolation in less memory, but quarter-width isolation beat it after the freeze; spiking encoder: fewer operations in an estimate, higher measured CPU cost
-  **Did not survive** the tested STDP rule vs a same-size random update; homeostasis with backprop; sleep vs replay at a matched budget; and the earlier findings on the 'didn't survive' slide
-  **Unresolved** Transformer working memory (benchmark validity not established); the SNN version of the isolation-width test

Thank you. Questions?

Backup: final accuracy and forgetting by method

	CNN, 3 tasks	SNN, 3 tasks	SNN+Transformer, 3 tasks	CNN, 5 tasks
Final ACC (mean $\pm$ SD)				
naive	0.555 $\pm$ 0.013	0.550 $\pm$ 0.130	0.452 $\pm$ 0.016	0.416 $\pm$ 0.098
EWC ($\lambda=100$)	0.674 $\pm$ 0.030	0.716 $\pm$ 0.199	–	–
replay	0.970 $\pm$ 0.009	0.814 $\pm$ 0.162	0.862 $\pm$ 0.038	0.926 $\pm$ 0.049
sleep	0.978 $\pm$ 0.001	0.889 $\pm$ 0.003	0.887 $\pm$ 0.045	0.928 $\pm$ 0.065
sleep, matched	–	–	0.879 $\pm$ 0.043	–
isolation	0.979 $\pm$ 0.001	0.833 $\pm$ 0.073	0.832 $\pm$ 0.037	0.976 $\pm$ 0.000
FORGET (mean)				
naive	0.631	0.321	0.536	0.612
replay	0.002	0.023	0.019	0.006
sleep	0.001	0.006	0.006	0.003
sleep, matched	–	–	0.020	–

Seeds: 3 per arm, except SNN+Transformer (replay 8, sleep 6, sleep matched 8). Paper Table 3.

Backup: prospectively specified local-plasticity contrasts

Contrast	Δ solve rate [95% CI]	Fisher p (Holm)	Δ AUC [95% CI]	Welch p (Holm)
STDP vs homeostasis alone	-0.20 [-0.51, +0.11]	0.474 (1.000)	-0.101 [-0.323, +0.121]	0.338 (0.724)
STDP vs matched random	+0.00 [-0.34, +0.34]	1.000 (1.000)	+0.033 [-0.260, +0.326]	0.818 (0.818)
homeostasis vs backprop	+0.20 [-0.11, +0.51]	0.474 (1.000)	+0.200 [+0.024, +0.377]	0.029 (0.117)
random vs homeostasis	-0.20 [-0.51, +0.11]	0.474 (1.000)	-0.134 [-0.372, +0.104]	0.241 (0.724)
DFA + homeostasis vs DFA	+0.50 [+0.07, +0.74]	0.070 (0.070)	+0.326 [+0.101, +0.551]	0.009 (0.017)

DoorKey-6x6, 10 seeds per arm. First four rows: one Holm family per statistic (backprop as the global signal). Last row: the DFA family (specified separately). Newcombe CIs for solve rates, Welch for AUC.

Backup: protocol constants

PPO	16 envs $\times$ 128 steps, 4 epochs, minibatch 256, clip 0.2, $\gamma = 0.99$, GAE $\lambda = 0.95$, KL stop 0.03
Learning rate	0.001 (CNN, SNN); 0.0003 (SNN+Transformer)
Frames per task	150k (CNN, SNN); 400k (SNN+Transformer); 450k (5-task SNN, fair budget)
SNN	4 time steps, learnable leak (init 0.9), surrogate slope 25, membrane reset each frame
Plasticity	$\alpha = 0.03$; homeostasis target 0.15, rate 0.01
Sleep	every 25 updates + task end; 700 steps per phase; 5,000 states per task
Evaluation	100 episodes per task (continual); 200 (single task)
Memory accounting	188 B/state (CNN, SNN); 632 B/state (SNN+Transformer)

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